

ISOLATION BY DISRUPTIVE SELECTION

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THOUGH speciation is one of the more striking features of evolution, direct experimental evidence concerning the origin of species is limited.

Artificial species have been made by allopolyploidy, but this, like any other process that involves the establishment of a new species from the hybrids between two pre-existing species, can only be secondary. The key process in speciation must involve

the splitting of one population into two mutually isolated populations.

Our present theories of such speciation, based on considerable taxonomic, biogeographical and ecological evidence, implicate divergence during a period of geographical isolation and selection against hybridization after the geographical isolation has broken down¹. Supporting evidence is available²⁻⁴ demon-

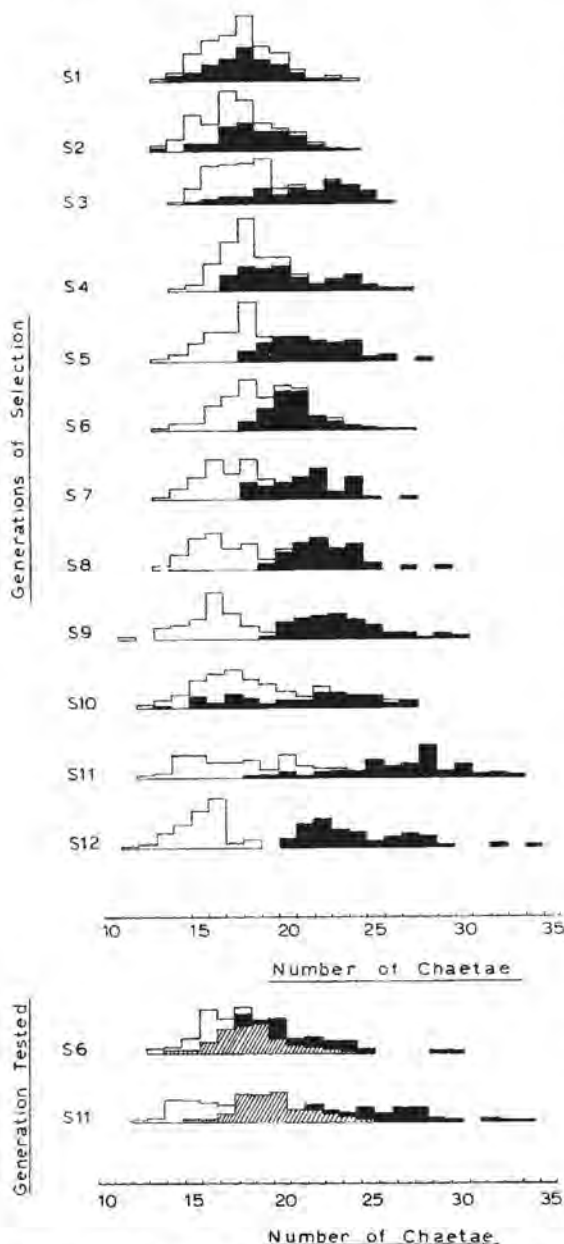


Fig. 1. Top, distribution curves of sternopleural chaeta number in each generation of the selection experiment. White: progeny of low females. Black: progeny of high females; Bottom, distribution curves obtained in the test assays of the progeny of S6 and S11 flies. White: progeny of low $\times$ low mating. Hatched: combined progenies of high $\times$ low and low $\times$ high matings (which do not differ). Black: progeny of high $\times$ high mating. These curves should be compared with those for S7 and S12 in Fig. 1 (top).

strating genetic variance affecting mating preference and showing that the frequency with which two genotypes form hybrids can be reduced by selection. However, the key demonstration that a single wild-type population can be converted by selection into two populations that are mutually isolated in the conditions in which they have to maintain themselves has not hitherto been made.

Mather⁵ not only argued that disruptive selection should be able to establish two or more forms in one interbreeding population, but also suggested that in

appropriate circumstances it should be able to promote isolating mechanisms. Thoday^{6,7}, Thoday and Boam⁸ and Millicent and Thoday⁹ have shown that polymorphisms are rather easily established and maintained by disruptive selection, but none of their experiments was of the kind in which isolation was to be expected.

The purpose of this article is to present a preliminary description of the results of an experiment in which a single wild-type population of *Drosophila melanogaster* has with astonishing rapidity been split into two populations that produce few hybrids in what to them is nature.

The experiment was started using the progenies of four wild females captured together in the same dustbin (garbage can) in an estate on the outskirts of Cambridge. The four progenies were combined as the 'Southacre' wild stock. Two generations later, 20 virgin females and 20 males from each of four cultures of this stock were assayed for sternopleural chaeta number, and from each sex the 8 flies with the highest and the 8 flies with the lowest chaeta numbers were selected regardless of the culture from which they came. (When one or two of several flies with the same chaeta number had to be selected, random numbers were used to ensure against bias towards choosing flies from particular cultures.)

These 32 selected flies were placed together in a single 3 in. $\times$ 1 in. vial and left there to mate for 24 hr. They were then re-etherized, the males were discarded and the females separated once more into the 8 high and 8 low according to their chaeta numbers. The 8 high females were divided at random into two lots of four and placed in two culture bottles. The 8 low females were likewise divided and placed in two other bottles. This process was repeated each generation thereafter.

Response to selection. The results are illustrated in Fig. 1 (top), which shows the distribution curves of chaeta number obtained in each generation, and distinguishes the progeny of the high females from the progeny of the low females. There was rapid divergence of the two halves of the population, as a result of which from generation 4 onward almost all the high flies selected came from the progeny of high females and almost all the low flies selected came from the progeny of low females. Apart from generation 10, in which circumstances were unusual, partly owing to electric power cuts which affected the culture room temperature and partly inasmuch as special arrangements, which involved a 24-hr. delay in mating, were made to accommodate the Christmas holiday, there was a steady and remarkably rapid approach to discontinuity of the distributions which in the latest generation do not overlap.

The mating system permits random mating and ensures that unless there are departures from random mating or unless 'hybrids' differ in viability from 'non-hybrids', half the flies in each half of the population will be 'hybrids' produced by the mating of high females with low males or of low females with high males. The rapid decrease in the overlap between the distribution curves and the discontinuity of the distribution curves in the latest generation clearly show that 'hybrids' must have occurred with very much lower frequencies than this, unless the hybrids are for some reason indistinguishable from one of the non-hybrid classes of fly.

Chaeta numbers of hybrids. In generations S6 and S11 additional samples of flies were taken with the aim of obtaining information about the chaeta

number distribution of hybrids. These additional samples were assayed and selected exactly as for the continuance of the line, but set up under a different mating system. Four of the high females were placed in one mating tube with 4 of the high males; the other 4 high females were placed in another tube with 4 of the low males. Four of the low females were placed with the remaining 4 high males, and the remaining 4 low females were placed with the remaining 4 low males. After 24 hr. the males were destroyed and the four lots of four females were placed in 4 separate culture bottles to give progeny from the four classes of mating high $\times$ high, high $\times$ low, low $\times$ high and low $\times$ low that would occur with equal frequency in random mating. Twenty of each sex of each of these four progenies were assayed and provided the distribution curves given in Fig. 1 (bottom) which are to be compared with those for generations 7 and 12 in Fig. 1 (top).

It is immediately evident that the hybrid modes are intermediate between those of the non-hybrids and hence that the discontinuity of the distributions for the last generation in Fig. 1 (top), and the decline of the overlap between the distributions in the later generations of the selection experiment were the result of a rapid decline in the frequency of 'hybrids' in the selected line. Whether this arises from pronounced mating preferences or from inability of the hybrid flies to compete as larvae in the same culture with the non-hybrids is not yet certain, but that hybrids have become very rare in the selection line is clear. The high and the low flies show a strong degree of reproductive isolation in the sense of Dobzhansky¹⁰.

The condition under which Mather⁶ proposed disruptive selection should give rise to polymorphism was that the two or more classes selected should be interdependent. This condition held in the earlier disruptive selection experiments in which genetic polymorphisms were produced and maintained⁶⁻⁹. Mather further argued that if the two classes that were selected could be independent the expected result would be reproductive isolation, assuming, of course, that suitable genetic variance were available.

The present experiment complements its predecessors, for by mating all the selected individuals together in the same vial the two selected classes have been given the opportunity to become inde-

pendent should they be able to do so. The result is as Mather predicted. The only surprise is the remarkable rapidity with which the result was produced, despite the fact that 20 per cent of the flies assayed were selected, a rather low selection pressure for an artificial selection experiment.

Thoday and Boam⁸ and Millicent and Thoday⁹ have pointed out that their demonstrations that divergence is possible without isolation have a bearing on our assessment of the possibility of sympatric speciation. Streams and Pimentel¹¹ have since obtained consonant results. The present experiment, in which divergence and reproductive isolation have originated without any geographic or analogous isolation when the flies were mating, can only add strength to this point of view.

We do not, however, wish it to be thought that we regard this as a demonstration that sympatric speciation occurs in Nature, for such a conclusion cannot be drawn from the results of laboratory selection experiments. The evidence that most speciation in Nature is allopatric is very strong¹, and it is for the student of natural populations to determine whether and how often sympatric speciation occurs. However, sympatric speciation has frequently been supposed impossible on theoretical grounds, and these can no longer be regarded as sound. Though selection pressures as high as those used in the present experiment may be uncommon in Nature, the speed with which the present results were obtained clearly forces us to consider the possibility that natural selection pressures might readily induce divergence and complete isolation in an appropriately heterogeneous habitat.

¹ Mayr, E., *Systematics and Origin of Species* (Columbia Univ. Press, 1942).

² Merrell, D. J., *Evolution*, **4**, 326 (1950).

³ Koopman K. F., *Evolution*, **4**, 135 (1950).

⁴ Knight, G. R., Robertson, A., and Waddington, C. H., *Evolution*, **10**, 14 (1956).

⁵ Mather, K., *Evolution*, **9**, 52 (1955).

⁶ Thoday, J. M., *Nature*, **181**, 1124 (1958).

⁷ Thoday, J. M., *Heredity*, **14**, 35 (1960).

⁸ Thoday, J. M., and Boam, T. B., *Heredity*, **13**, 205 (1959).

⁹ Millicent, E., and Thoday, J. M., *Heredity*, **18**, 199 (1961).

¹⁰ Dobzhansky, Th., *Genetics and the Origin of Species* (Columbia Univ. Press, 1942).

¹¹ Streams, F. A., and Pimentel, D., *American Naturalist*, **95**, 201 (1961).